> The cleantech category shows up frequently among the most-funded startups that later failed. Why might this be the case? Is cleantech inherently risky, or are there external factors that make success harder in this space?

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Cleantech is not inherently doomed, but it is unusually difficult to finance and commercialize—especially under the traditional venture-capital model.

Why heavily funded cleantech startups can still fail

1. **They require far more capital and time than typical software startups.**
   Cleantech companies often must develop physical technologies, build pilot facilities, obtain certifications, construct manufacturing capacity, and deploy infrastructure before generating meaningful revenue. These activities can require hundreds of millions of dollars and long development cycles. By contrast, software companies can often launch with relatively modest capital and scale through digital distribution. Cleantech therefore faces a longer and deeper "valley of death" between technical proof and commercial viability brookings.edu +1 .
2. **Technical success does not guarantee economic success.**
   A technology may work in a laboratory or pilot project but fail to achieve competitive costs at commercial scale. Manufacturing yields, durability, maintenance, supply chains, materials availability, and installation costs can all undermine the business case. Scaling physical production is also difficult because each additional unit generally requires additional equipment and capital, whereas software can often serve more customers at relatively low marginal cost energybalancesh...bstack.com.
3. **Cleantech demand depends heavily on external markets and policy.**
   Many projects compete with established energy technologies whose prices can change sharply. For example, Brookings attributes declining cleantech investment after 2011 partly to cheap natural gas, which made some solar and wind projects less economical, as well as to the struggles of highly funded firms such as Solyndra and Evergreen Solar brookings.edu. Changes in subsidies, tax credits, regulation, permitting, grid policy, or carbon prices can therefore materially alter a startup's market after it has already invested heavily.
4. **Commodity-price competition is especially damaging.**
   Cleantech firms may be competing with conventional energy sources, not merely with another startup. If oil, gas, or electricity prices move against them, customers may delay adoption even when the technology provides environmental benefits. Solar manufacturing illustrates this problem: commoditization reduced the attractiveness of investing in new panel-manufacturing innovations, while capital shifted toward lower-risk financing, installation, and service businesses brookings.edu.
5. **The financing structure may not match the business.**
   Venture capital generally seeks very large returns within a limited fund life, often requiring an acquisition or IPO within roughly five to ten years energybalancesh...bstack.com. Cleantech firms may need several rounds of funding before reaching commercial scale, and later rounds can dilute earlier investors or shift control to infrastructure funds, strategic corporations, or private-equity investors. A company can therefore be technologically promising but unable to raise the next round on acceptable terms.

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6. **Exit opportunities are less predictable.**
   Software startups often have clearer acquisition markets and more standardized pathways to rapid growth. Cleantech companies may have fewer potential acquirers, particularly when their assets are capital intensive, regulated, geographically constrained, or dependent on project finance. Research summarized by Brookings identifies the absence of an established exit mechanism as one factor making cleantech less suitable for conventional VC investing [brookings.edu].
7. **Investors may fund the wrong kinds of cleantech companies.**
   When investors become cautious, capital tends to flow toward relatively capital-efficient businesses—such as energy-efficiency software, transportation platforms, or solar financing—rather than deep-technology companies developing nuclear systems, new fuels, offshore wind, advanced materials, or breakthrough storage. Brookings found that 87% of U.S. cleantech VC dollars in 2016 went to late-stage companies, while early-stage funding had fallen to 13% [brookings.edu]. This can leave genuinely transformative technologies underfunded at the stage when they most need patient capital.

Why the failures are so visible

The observation that cleantech appears frequently among highly funded failures can be affected by **selection and visibility effects**:

- Cleantech startups often require unusually large funding rounds, so their failures attract substantial media and investor attention.
- The sector experienced periods of enthusiasm and overinvestment, followed by abrupt pullbacks.
- Large funding can create high expectations and encourage rapid expansion before the technology, unit economics, supply chain, or market is ready.
- The relevant comparison is not simply whether a company failed, but whether its risk, time horizon, capital requirements, and expected return were compatible with its financing model.

Indeed, Brookings reports that $25 billion invested in cleantech between 2006 and 2011 produced losses of more than half for investors in the cited MIT analysis—an outcome substantially worse than the performance of software or medical technology in that comparison [brookings.edu]. That does not mean every cleantech segment performs poorly; rather, it suggests that the conventional VC model is often a poor fit for capital-intensive clean-energy innovation.

Bottom line

Cleantech is risky because it combines **technology risk, scale-up risk, market risk, policy risk, commodity-price risk, and financing risk**. Many of these risks are external to the startup itself. The sector generally needs a broader financing mix: grants and public R&D support for early technical risk, strategic corporate investment, project finance and debt for infrastructure, and patient long-term equity for commercialization. Traditional VC can work well for software-like or highly scalable cleantech businesses, but it is often insufficient—or poorly matched—for physical technologies that require long development timelines and large amounts of capital [brookings.edu +1].

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